

LIAQUAT COLLEGE OF NURSING FEMALE, JAMSHORO

KNOWLEDGE, PRACTICES, AND CHALLENGES OF INFECTION PREVENTION AND CONTROL AMONG NURSES IN TERTIARY HEALTHCARE SETTINGS

A Descriptive Cross-Sectional Study

SUBMITTED BY

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DECLARATION

I, **Tasleem d/o Mohammed Ali Rind**, hereby declare that the dissertation topic *"Knowledge, Practices, and Challenges of Infection Prevention and Control Among Nurses in Tertiary Healthcare Settings"*, which I submitted is entirely the result of my own work. Every effort is made to indicate this clearly, with due reference to the literature. The contents of the thesis, in full or parts, have not been submitted to any other Institute or University for the award of any other degree or diploma.

The work was done under the supervision and guidance of **Sir Gobind Rai**.

Place: Liaquat College Of Nursing Female, Jamshoro

Date: _____

Name: TASLEEM

Signature: _____

CERTIFICATE

This is to certify that **Tasleem** has completed this research work titled *"Knowledge, Practices, and Challenges of Infection Prevention and Control Among Nurses in Tertiary Healthcare Settings"* as a research scholar under my supervision and guidance for the partial fulfillment of the requirement of the regulations laid down for the degree of Bachelor of Science in Nursing at Liaquat College Of Nursing Female, Jamshoro.

Supervisor's Name:

Gobind Rai

Signature: _____

DEDICATION

*In the name of **ALMIGHTY ALLAH**, the Most Gracious, the Most Merciful, whose infinite blessings and divine guidance illuminated every step of this journey.*

*To my beloved **parents**, whose endless sacrifices, heartfelt prayers, and unconditional love have been the foundation of everything I have achieved. Whatever I am today is because of you.*

*And to my beloved fiancé, **Afzal Khan**, whose unwavering love, patience, and constant encouragement gave me the strength to persevere through every challenge.*

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I owe a deep debt of gratitude to my loving **parents**, whose prayers, sacrifices, and unwavering moral support have been my greatest strength. I am equally grateful to all the **participants** and the nursing staff of LUMHS Jamshoro and Hyderabad campuses for their voluntary participation and cooperation during data collection, without which this study would not have been possible.

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ABSTRACT

Background: Hospital-acquired infections remain a significant challenge in healthcare settings worldwide. Nurses, being at the frontline of patient care, play a crucial role in infection prevention and control (IPC). However, studies from Pakistan consistently report a gap between knowledge and practice of IPC measures.

Objective: To assess the knowledge, practices, and institutional barriers related to infection prevention and control among nurses working at Liaquat University of Medical & Health Sciences (LUMHS), Jamshoro and Hyderabad.

Methods: A descriptive cross-sectional study was conducted among 103 nurses at LUMHS using a structured questionnaire with 15 items covering knowledge, practice, and barriers. Data were analyzed using IBM SPSS Version 23 with descriptive statistics and chi-square tests.

Results: The majority of participants were female (61.2%), aged 26-30 years (46.6%), and held BSN qualifications (65.0%). The mean knowledge score was 3.61/5 (SD=1.02) with 60.2% demonstrating good knowledge. The mean practice score was 7.31/9 (SD=1.28) with 73.8% reporting good practice. Critical gaps included: 50.5% incorrectly believing clean water alone destroys microorganisms, 30.1% handling body fluids without gloves, and 28.2% not attending annual IPC training. No significant gender or age associations were found ($p>0.05$).

Conclusion: Nurses demonstrated adequate overall knowledge but inconsistent practice implementation. The primary barriers are institutional rather than individual, suggesting that systemic interventions targeting training frequency, guideline accessibility, and PPE supply would be more effective than knowledge-based education alone.

Keywords: Infection control, nurses, healthcare-associated infections, hand hygiene, personal protective equipment, nosocomial infections, tertiary hospital, Pakistan

TABLE OF CONTENTS

Abstract

1. Introduction

- 1.1 Gaps in Current Research
- 1.2 Study Objectives

2. Review of Related Literature

- 2.1 Synthesis of Evidence

3. Materials and Methods

- 3.1 Study Design
- 3.2 Study Setting and Duration
- 3.3 Sample Size Calculation
- 3.4 Sampling Method
- 3.5 Inclusion and Exclusion Criteria
- 3.6 Data Collection Instrument
- 3.7 Data Collection Procedure
- 3.8 Ethical Considerations
- 3.9 Data Analysis

4. Results

- 4.1 Demographic Characteristics
- 4.2 Knowledge Assessment
- 4.3 Practice Assessment
- 4.4 Barriers to Infection Control
- 4.5 Overall Knowledge and Practice Scores
- 4.6 Chi-Square Analysis

5. Discussion

- 5.1 Knowledge Assessment
- 5.2 Practice Assessment
- 5.3 Barriers Analysis
- 5.4 Comparison with International Literature
- 5.5 Limitations

6. Conclusion

- 6.1 Recommendations

References

1. INTRODUCTION

Hospital-acquired infections (HAIs) remain a significant challenge in healthcare systems worldwide. Patients sometimes contract new infections during treatment, leading to worse outcomes, prolonged hospital stays, and elevated healthcare costs. These infections affect both patients and healthcare workers, particularly nurses who maintain the closest and most continuous patient contact.[1] In tertiary care hospitals handling complex cases, the presence of immunocompromised patients, invasive devices, and high patient turnover create environments where infectious agents spread more easily.[2]

Standard infection control measures address these risks effectively when properly implemented. Hand hygiene remains the cornerstone of prevention, recognized by the World Health Organization (WHO) as the single most effective measure to prevent HAIs.[3] Additional measures include proper use of personal protective equipment (PPE), safe needle and sharps disposal, regular environmental cleaning, and equipment sterilization.[4] International health organizations have published comprehensive guidelines for these practices. Yet compliance among healthcare workers, particularly in resource-limited settings, often falls short of recommended standards.[5]

Studies consistently document a persistent gap between theoretical knowledge and actual practice of infection control measures.[1,4,6] Nurses might correctly answer questions about protocols while admitting they skip proper technique during busy shifts. Several factors contribute to this gap, including lack of continuous education, inadequate training, insufficient resources, and excessive workload.[7,8]

Pakistan's healthcare system faces pressures that compound compliance challenges. Public hospitals operate with nurse-to-patient ratios below recommended levels. Supply issues leave wards without adequate protective equipment. Administrative oversight remains inconsistent.[2,5] A nurse working a 12-hour shift with twice the recommended patient load, facing glove shortages, and receiving no supervisory feedback exists in a markedly different context than the controlled settings where international protocols were developed. Tertiary hospitals treating the sickest patients represent settings where prevention failures carry the highest stakes, yet research examining these facilities in Pakistan remains limited.

1.1 Gaps in Current Research

Existing studies from Pakistan document the knowledge-practice disconnect, but important questions remain unanswered. Most research measures knowledge and practice separately without examining what prevents nurses from acting on their knowledge.[1,2,3,4,5] Survey methods dominate, creating potential for social desirability bias. Few studies employ observational methods or examine actual compliance through direct monitoring.

The specific context of tertiary hospitals has received insufficient attention. These facilities handle referral cases, treat complex conditions, and maintain intensive care units where infection risks are elevated. Whether barriers differ between tertiary and district hospitals remains unclear. Most

importantly, studies treat structural constraints as background factors rather than central inquiry subjects. When international guidelines assume consistent access to running water and unlimited sterile gloves, how do nurses in resource-limited settings prioritize competing demands?

1.2 Study Objectives

This research examines infection control among nurses in tertiary care settings through four objectives:

1. Assess current knowledge levels regarding infection prevention protocols
2. Evaluate reported infection control practices during patient care
3. Identify institutional, environmental, and individual barriers impeding adherence
4. Analyze relationships between demographics, knowledge, practices, and barriers

2. REVIEW OF RELATED LITERATURE

Khowaja and colleagues[1] conducted a cross-sectional study at Liaquat Institute of Medical & Health Sciences Hospital, Thatta, with 30 nurses. The study found that nurses demonstrated satisfactory understanding of infection control principles, but actual compliance fell below acceptable standards. Inadequate continuing education and irregular protective equipment availability were identified as key obstacles. However, the study did not explore how these barriers interact: whether guaranteed glove supply alone would improve compliance, or whether multiple factors need simultaneous change.

Batool and colleagues[2] studied 138 nurses at Sandeman Provincial Civil Hospital and Bolan Medical Complex Hospital in Quetta. They documented concerning practices including insufficient hand hygiene and found that 99.3% of participants reported using gloves and 97.8% used masks. As per perceptions, 66.7% agreed that nurses with common colds should be excused from duty, and 76.8% acknowledged that cultural attitudes influence infection spread. Environmental factors emerged as significant contributors, particularly poor sanitation and absent monitoring. Infection control practices were statistically non-significant according to age and work experience ($p>0.05$).

Kousar's team[3] studied nurses at Aziz Bhatti Shaheed Hospital, Gujrat, reporting high awareness but inconsistent implementation. Utilizing the Health Belief Model as a theoretical framework, their analysis emphasized the role of infection control committees and surveillance programs. The chi-square analysis revealed significant gender differences in knowledge, with female nurses showing higher knowledge regarding antiseptic use, autoclaving, and hand washing practices. However, the study did not examine whether institutional oversight systems actually function in Pakistani hospitals.

Soomro and colleagues[4] examined Isra University Hospital, Hyderabad, with 60 nurses, focusing on standard infection control precautions. They found high adherence to basic measures (85% for hand hygiene and standard precautions) but lower compliance with specialized procedures-only 56.7% wore gowns and gloves when contact precautions were required, and only 45% consistently wore gloves during blood collection. These findings highlight that while general preventive measures are maintained, specific high-risk procedures require additional attention.

Baqar and colleagues[5] documented occupational hazards from blood exposure among 111 nursing staff at a tertiary care hospital in Khairpur. Using a WHO-validated questionnaire combined with direct observation, they found 88.3% agreed healthcare workers should be vaccinated, but 66.7% of nurses recapped needles despite knowing the associated dangers. The study uniquely incorporated observation alongside self-reporting, revealing that 0.9% compliance with hand washing before aseptic tasks and 19.8% after glove removal-dramatically lower than self-reported rates. Gender differences were significant ($p=0.000$), with female nurses showing higher compliance with hand hygiene and PPE use.

2.1 Synthesis of Evidence

Common themes emerge across these five studies. Knowledge levels appear generally adequate-most nurses can identify proper procedures when asked. Yet practice consistently falls short of knowledge, a pattern that holds across different cities and hospital types. The explanations converge on resource constraints, workload pressures, and training gaps. However, current research describes barriers without deeply analyzing their relative importance or interaction effects. Moreover, most studies rely on self-reported data, which likely inflates compliance estimates, as demonstrated by the dramatic discrepancy between self-reported and observed compliance in the Baqar study.[5]

The present study addresses these limitations by examining knowledge, practices, and barriers together within the tertiary hospital context at LUMHS, using a comprehensive instrument that captures institutional barriers alongside individual knowledge and practice measures.

3. MATERIALS AND METHODS

3.1 Study Design

A descriptive cross-sectional study was conducted to assess the knowledge, practices, and challenges of infection prevention and control among nurses working at tertiary healthcare settings.

3.2 Study Setting and Duration

The study was carried out at Liaquat University of Medical & Health Sciences (LUMHS), including the main campus hospital in Jamshoro and its branch facility in Hyderabad, Sindh, Pakistan. Data collection was conducted from January to February 2026. LUMHS is a major public sector tertiary care teaching hospital providing specialized medical services across multiple departments including Medicine, Surgery, Pediatrics, ICU, CCU, Emergency, Orthopedics, Cardiology, Pulmonology, Urology, Nephrology, Gynecology, and Operation Theater.

3.3 Sample Size Calculation

The sample size was calculated using Slovin's formula:

$$n = N / (1 + N \times e^2)$$

Where N = total nursing population at LUMHS (~140 nurses across both campuses) and e = margin of error (0.05). This yielded n = 140 / (1 + 140 × 0.0025) = 140 / 1.35 ≈ **103 participants**.

3.4 Sampling Method

Convenience (non-probability) sampling method was adopted. Nurses available and willing to participate during the data collection period were enrolled from various clinical departments across both campuses.

3.5 Inclusion and Exclusion Criteria

Inclusion: All registered staff nurses (male and female) currently employed at LUMHS in direct patient care roles. **Exclusion:** Student nurses, interns, nurses on leave during data collection, those who declined participation, and administrative nursing staff not involved in direct patient care.

3.6 Data Collection Instrument

Data were collected using a structured, self-administered questionnaire designed based on previous published literature and WHO infection control guidelines.[5,9] The questionnaire consisted of two sections:

Section A - Demographic Data (5 items): Gender, age, marital status, professional qualification, and department.

Section B - Knowledge, Practice, and Barriers (15 items): Questions were categorized into three overlapping domains: Knowledge (5 items covering HAI transmission, nosocomial infection definition, WHO hand hygiene moments, universal precautions, and water disinfection); Practice (9

items covering training availability, hand washing, PPE use, body fluid handling, and waste management); and Barriers (5 items covering training frequency, guideline accessibility, knowledge adequacy, PPE availability, and workshop attendance). Response options were Yes/No/Don't Know for all items.

3.7 Data Collection Procedure

The questionnaire was physically distributed to nurses during their duty shifts across different departments at both LUMHS campuses. The purpose of the study was explained to each participant. Participants were given adequate time to complete the questionnaire and assistance was provided for ambiguous items. Completed questionnaires were collected on the same day.

3.8 Ethical Considerations

Verbal and written informed consent was obtained from all participants. Participation was voluntary, and nurses could withdraw at any time without consequence. All responses were anonymous and confidential. The study adhered to the principles of the Declaration of Helsinki. Approval was obtained from hospital administration.

3.9 Data Analysis

Data were entered and analyzed using IBM SPSS Statistics Version 23. Descriptive statistics (frequencies, percentages, mean, standard deviation) were used to summarize demographic characteristics and response distributions. Chi-square test (χ^2) was used to test associations between demographic variables and knowledge/practice scores. Cross-tabulation was used to compare responses across demographics. A p-value ≤ 0.05 was considered statistically significant.

4. RESULTS

4.1 Demographic Characteristics

A total of 103 nurses participated in this study. Table 1 presents the demographic profile of participants. The majority were female (61.2%), aged 26-30 years (46.6%), married (53.4%), and held BSN qualifications (67.0%). Nurses were distributed across multiple clinical departments as shown in Figure 4.

Table 1: Demographic characteristics of study participants including gender, age, marital status, and professional qualification distribution among nurses at LUMHS tertiary care hospital (n=103)

Variable	Category	Frequency (n)	Percentage (%)
Gender	Male	40	38.8%
	Female	63	61.2%
Age (Years)	≤25	20	19.4%
	26-30	48	46.6%
	31-40	24	23.3%
	Above 40	11	10.7%
Marital Status	Single	44	42.7%
	Married	55	53.4%
	Widowed	4	3.9%
Qualification	Diploma in Nursing	29	28.2%
	BSN	69	67.0%
	Post RN BSCN	5	4.9%

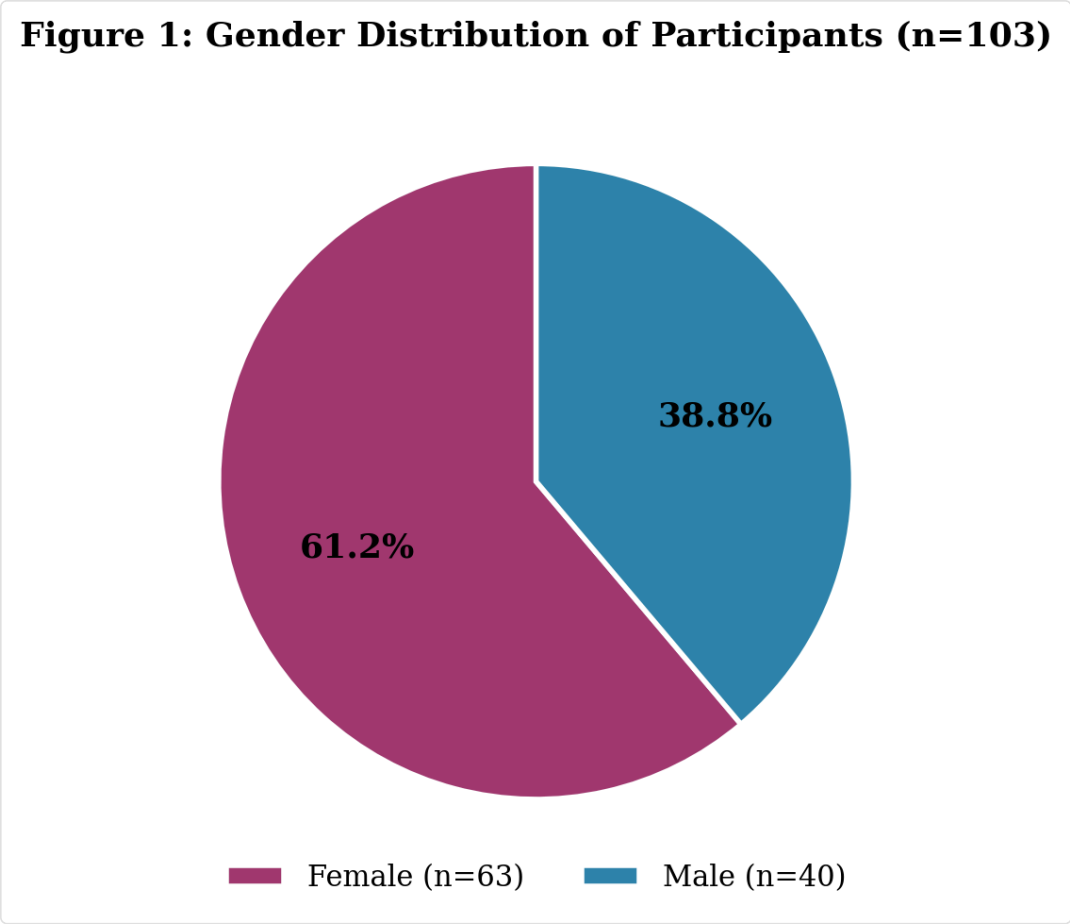


Figure 1: Pie chart showing gender distribution of study participants. Female nurses constituted the majority (61.2%, n=63) compared to male nurses (38.8%, n=40), reflecting the gender composition typical of the nursing profession at LUMHS tertiary care hospital (n=103)

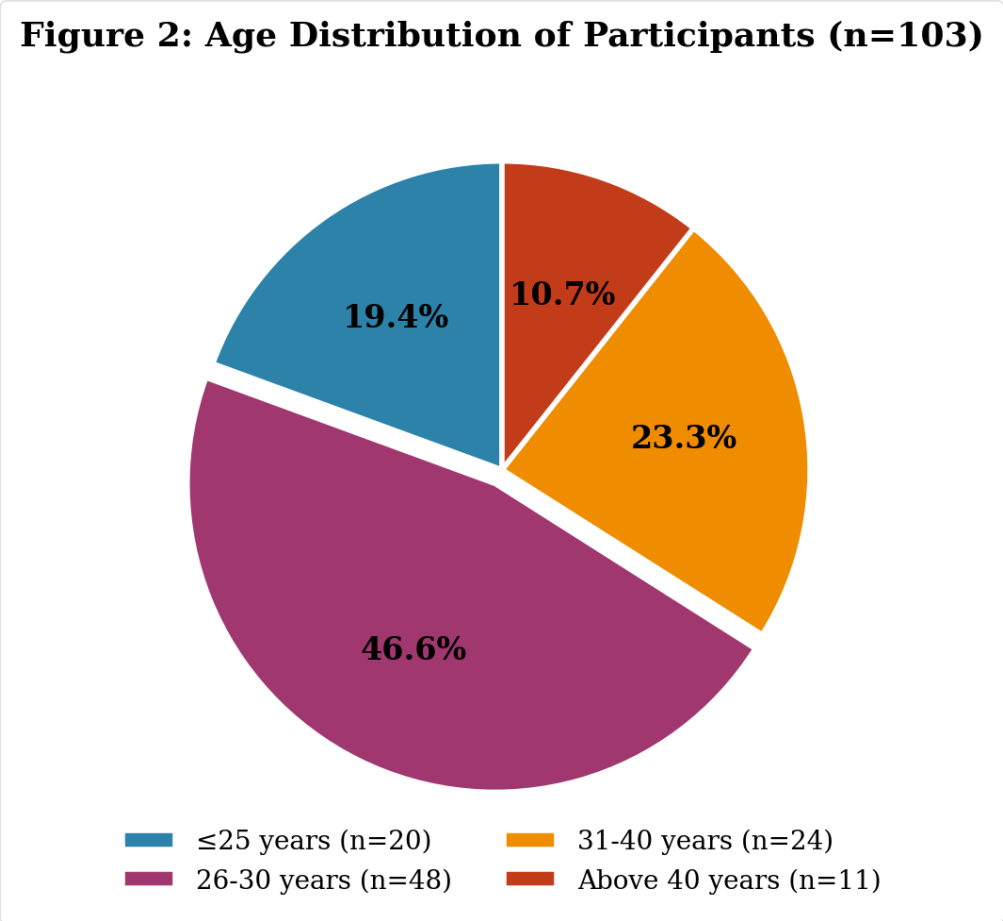


Figure 2: Pie chart showing age distribution of participants. The largest age group was 26-30 years (46.6%, n=48), followed by 31-40 years (23.3%, n=24), indicating a predominantly young nursing workforce at LUMHS (n=103)

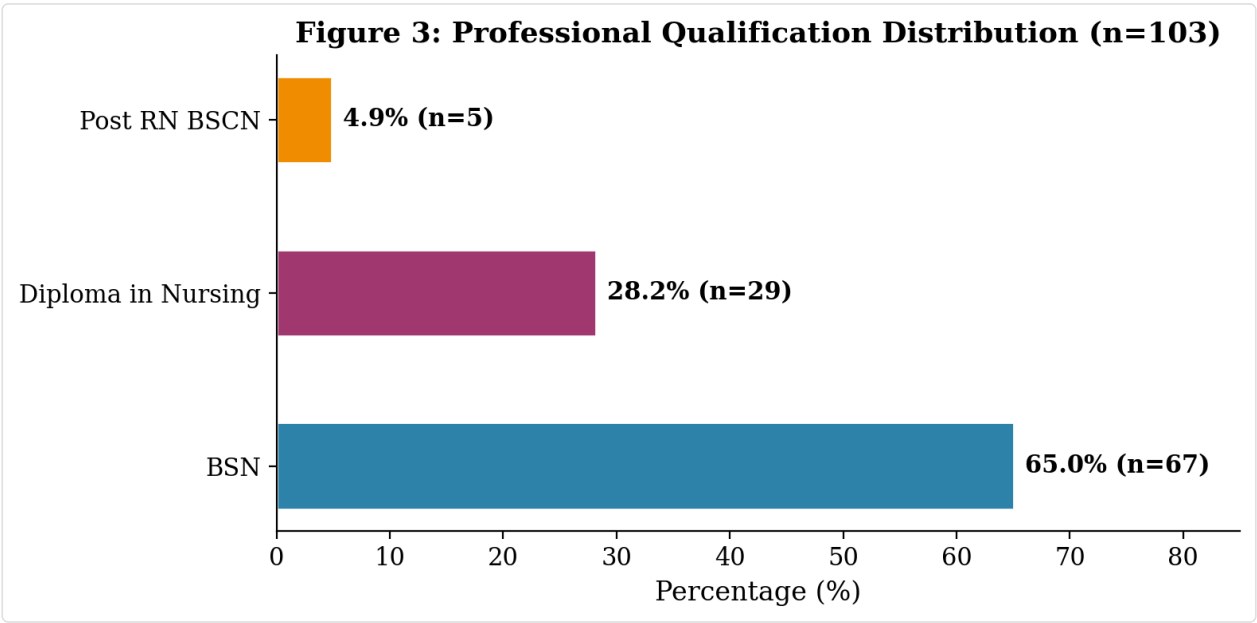


Figure 3: Horizontal bar chart showing professional qualification distribution. BSN was the most common qualification (67.0%, n=69), followed by Diploma in Nursing (28.2%, n=29) and Post RN BSCN (4.9%, n=5) (n=103)

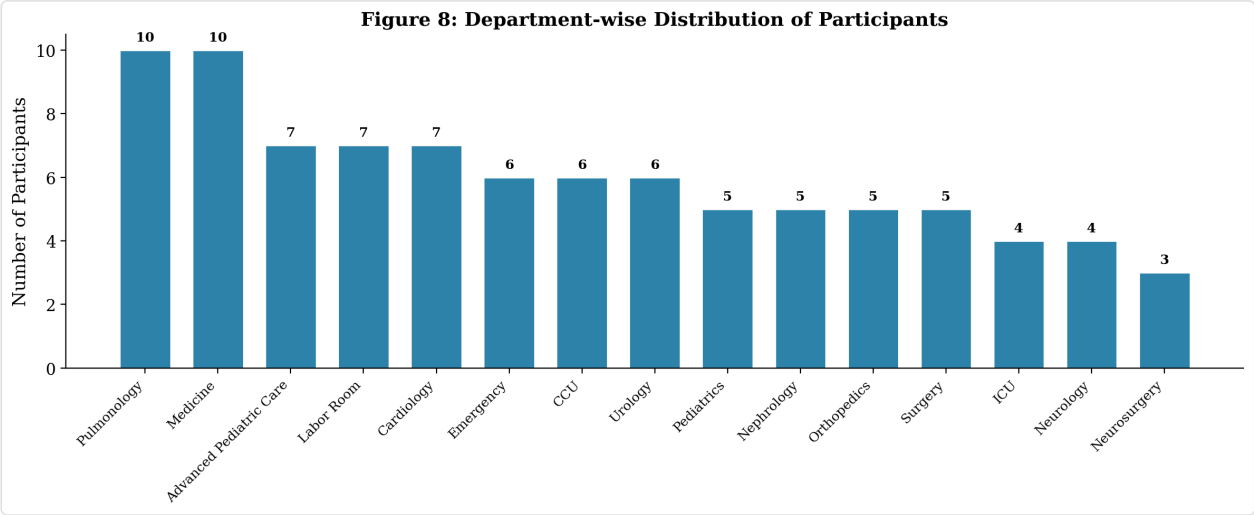


Figure 4: Bar chart showing department-wise distribution of participating nurses across clinical departments at LUMHS Jamshoro and Hyderabad campuses, illustrating the variety of clinical settings represented in the study (n=103)

4.2 Knowledge Assessment

Table 2 presents the responses to knowledge-related items. The majority of nurses (92.2%) correctly identified that hospital-acquired infections can be transmitted through contaminated medical equipment, and 78.6% were aware of the WHO Five Moments of Hand Hygiene. However, a critical knowledge gap was identified: **50.5% incorrectly believed that clean water alone is sufficient to destroy harmful microorganisms**. Additionally, 25.2% could not correctly define nosocomial infections.

Table 2: Distribution of responses to knowledge assessment items on infection prevention and control, showing frequency and percentage of Yes, No, and Don't Know responses among nurses at LUMHS (n=103)

Knowledge Item	Yes n(%)	No n(%)	Don't Know n(%)
Q3. HAIs can be transmitted through contaminated medical equipment	95 (92.2%)	6 (5.8%)	2 (1.9%)
Q4. Nosocomial infection refers to infection acquired during hospital stay	72 (69.9%)	26 (25.2%)	5 (4.9%)
Q6. Aware of WHO Five Moments of Hand Hygiene	81 (78.6%)	22 (21.4%)	0 (0.0%)
Q7. All HCWs and patients should be considered potentially infectious	74 (71.8%)	25 (24.3%)	4 (3.9%)
Q8. Clean water alone is sufficient to destroy harmful microorganisms*	52 (50.5%)	48 (46.6%)	3 (2.9%)

**Correct answer is "No" - only 46.6% answered correctly. This represents a critical knowledge gap.*

The overall mean knowledge score was **3.61 out of 5** (SD = 1.02). Of the 103 participants, 62 (60.2%) demonstrated good knowledge (score ≥4/5) and 41 (39.8%) had poor knowledge (score <4/5).

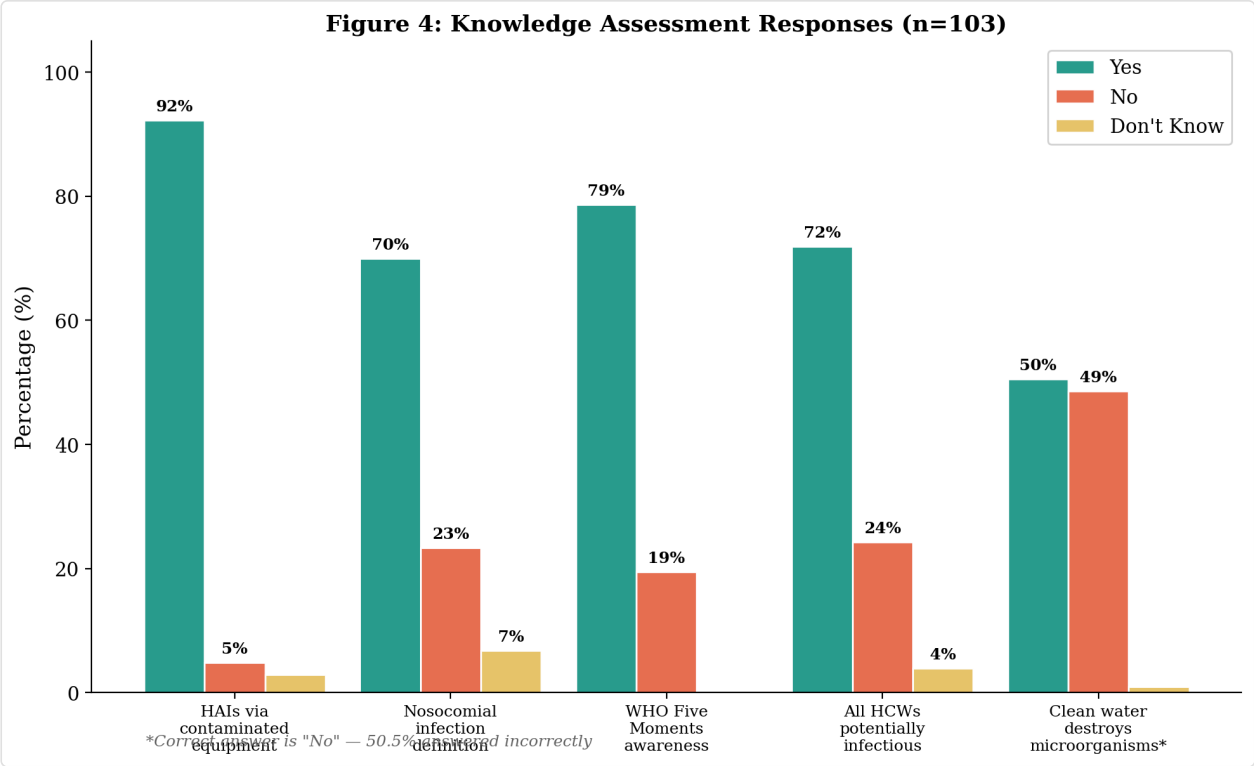


Figure 5: Grouped bar chart showing response distribution for five knowledge assessment items. Notably, 50.5% of nurses incorrectly believed clean water alone destroys microorganisms (marked with asterisk), representing the most critical knowledge gap identified in the study (n=103)

4.3 Practice Assessment

Table 3 presents the self-reported infection control practices. Self-reported hand hygiene compliance was high at 88.3%. Nearly all nurses (98.1%) agreed that proper IPC practices improve patient outcomes. However, concerning gaps were noted: **30.1% admitted to handling body fluids with bare hands** when gloves are unavailable, and 28.2% did not attend IPC training or workshops even once a year.

Table 3: Distribution of self-reported infection control practice responses among nurses, showing compliance levels for each practice item including training participation, hand hygiene, PPE use, and waste handling (n=103)

Practice Item	Yes n(%)	No n(%)	Don't Know n(%)
Q1. Periodic IPC training conducted in department	83 (80.6%)	17 (16.5%)	3 (2.9%)
Q2. IC guidelines available and accessible	77 (74.8%)	26 (25.2%)	0 (0.0%)
Q5. Hand washing before/after every patient contact	91 (88.3%)	9 (8.7%)	3 (2.9%)
Q9. Body fluids never handled with bare hands	68 (66.0%)	31 (30.1%)	4 (3.9%)
Q11. PPE always available when needed	89 (86.4%)	14 (13.6%)	0 (0.0%)
Q12. Masks/eye protection during invasive procedures	88 (85.4%)	15 (14.6%)	0 (0.0%)
Q13. Attend IPC training at least once/year	74 (71.8%)	29 (28.2%)	0 (0.0%)
Q14. PPE when handling contaminated waste	82 (79.6%)	21 (20.4%)	0 (0.0%)
Q15. Proper IPC improves patient outcomes	101 (98.1%)	0 (0.0%)	2 (1.9%)

The overall mean practice score was **7.31 out of 9** (SD = 1.28). Of the participants, 76 (73.8%) demonstrated good practice (score ≥7/9) and 27 (26.2%) had poor practice (score <7/9).

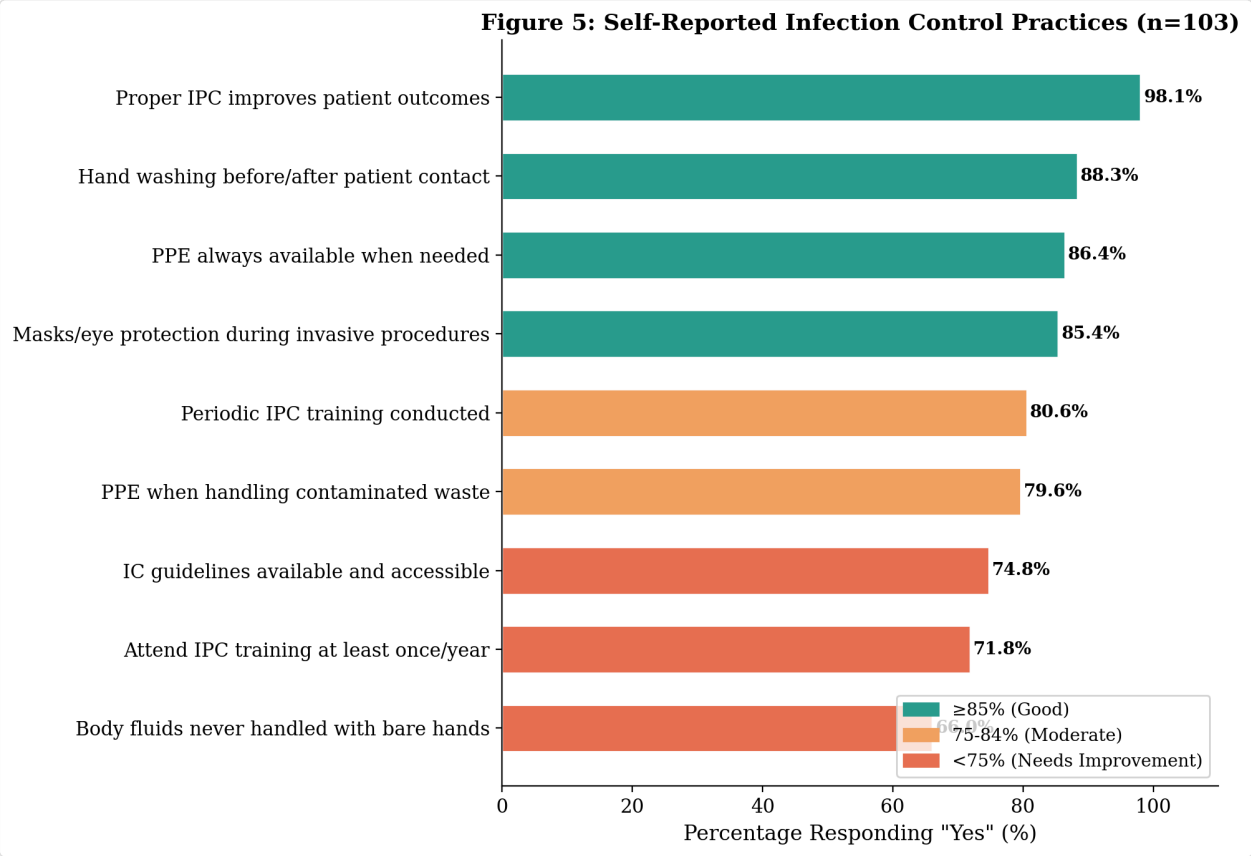


Figure 6: Horizontal bar chart showing self-reported infection control practices ranked by compliance level. Items are color-coded: green (good, above 85%), orange (moderate, 75-84%), and red (needs improvement, below 75%). Body fluid handling without gloves (66.0%) was the lowest-scoring practice item (n=103)

4.4 Barriers to Infection Control

Table 4 presents the identified barriers based on responses indicating lack of institutional support. The most commonly reported barrier was **lack of annual IPC training and workshops (28.2%)**, followed by inaccessibility of infection control guidelines (25.2%). Despite LUMHS being a tertiary teaching hospital, one in four nurses reported that IC guidelines were not readily available in their department.

Table 4: Institutional barriers to infection control as identified by nurses, showing percentage reporting each barrier based on negative responses to relevant questionnaire items (n=103)

Barrier	% Reporting Barrier ("No")
Lack of annual IPC training/workshops	**28.2%**
IC guidelines NOT available/accessible	**25.2%**
Inadequate personal knowledge of HAI prevention	20.4%
Periodic IPC training NOT conducted	16.5%
PPE NOT always available when needed	13.6%

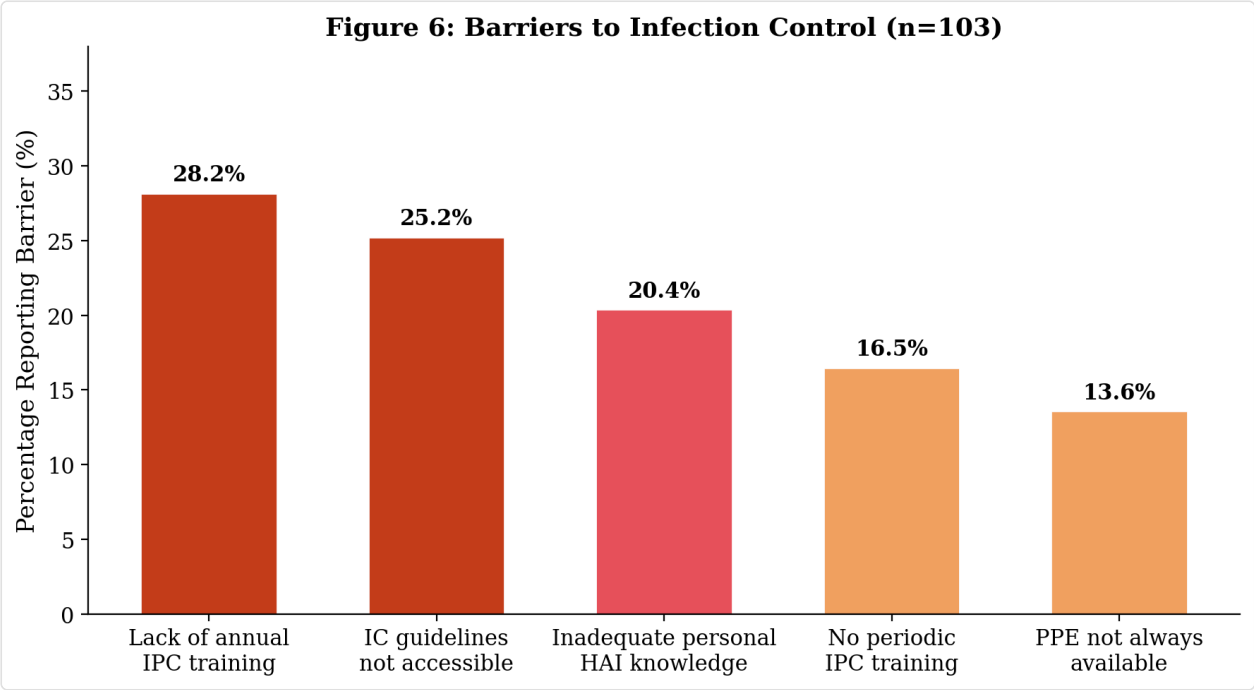


Figure 7: Bar chart showing the five main institutional barriers to infection control identified by nurses. Lack of annual IPC training (28.2%) and inaccessible IC guidelines (25.2%) were the most frequently reported barriers, highlighting systemic gaps in institutional support (n=103)

4.5 Overall Knowledge and Practice Scores

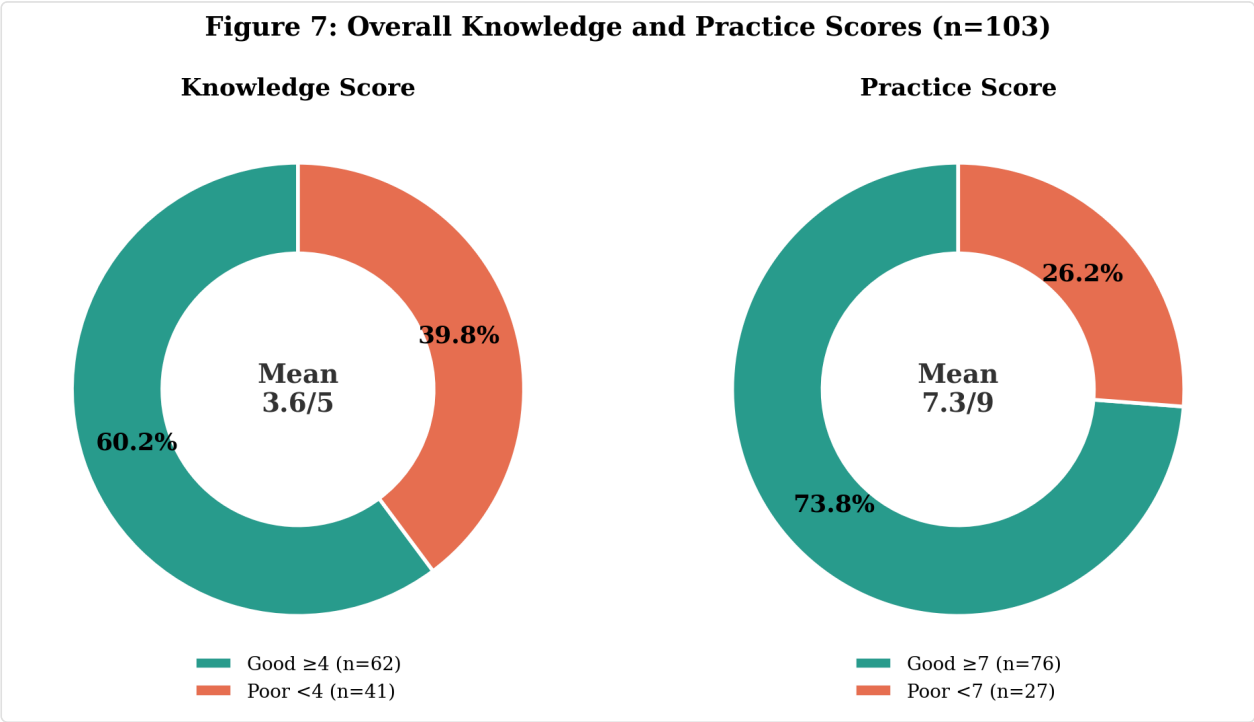


Figure 8: Donut charts comparing overall knowledge and practice scores. Knowledge: 60.2% scored good (4 or above out of 5, mean=3.6). Practice: 73.8% scored good (7 or above out of 9, mean=7.3). The higher self-reported practice scores may reflect social desirability bias (n=103)

Key Score Summary

- **Knowledge Score:** Mean = 3.61/5 (SD = 1.02) - Good knowledge: 60.2%, Poor: 39.8%
- **Practice Score:** Mean = 7.31/9 (SD = 1.28) - Good practice: 73.8%, Poor: 26.2%
- Practice scores were higher than knowledge scores, likely due to self-reporting bias

4.6 Association Between Demographics and Responses

Gender: No statistically significant associations were found between gender and any knowledge or practice variable (all $p > 0.05$), as shown in Table 5. However, a near-significant trend was observed for body fluid handling ($\chi^2 = 3.051$, $p = 0.081$), where male nurses reported higher compliance (77.5% vs. 58.7%).

Table 5: Association between gender and key knowledge/practice responses using Pearson Chi-Square test with Yates' correction. Statistical significance set at $p \leq 0.05$ (n=103; Male=40, Female=63)

Item	Male Yes%	Female Yes%	χ^2	p-value	Sig
HAIs via contaminated equipment	97.5%	88.9%	1.473	0.225	NS
Hand washing compliance	90.0%	87.3%	0.010	0.920	NS
Body fluids without bare hands	77.5%	58.7%	3.051	0.081	NS
WHO Five Moments awareness	77.5%	79.4%	0.000	1.000	NS
PPE always available	90.0%	84.1%	0.305	0.581	NS
Attend annual training	77.5%	68.3%	0.627	0.428	NS
IPC improves outcomes	97.5%	98.4%	0.000	1.000	NS

NS = Not Significant ($p > 0.05$). Yates' correction applied.

Age: No statistically significant associations were found between age groups and responses (all $p > 0.05$), as shown in Table 6. Self-assessed adequate knowledge showed a near-significant trend ($\chi^2 = 6.557$, $p = 0.088$), with younger nurses (≤25 years: 95.0%) reporting higher confidence than those aged 31–40 (62.5%).

Table 6: Association between age groups and key knowledge/practice responses using Pearson Chi-Square test. Four age categories were compared across selected questionnaire items (n=103)

Item	≤25	26-30	31-40	>40	χ^2	p-value	Sig
Adequate HAI knowledge	95.0%	77.1%	62.5%	72.7%	6.557	0.088	NS
PPE available	85.0%	93.8%	75.0%	81.8%	5.094	0.165	NS
Masks during procedures	95.0%	85.4%	75.0%	90.9%	3.836	0.280	NS
Hand washing	90.0%	83.3%	91.7%	100%	2.933	0.402	NS
Nosocomial definition	75.0%	66.7%	83.3%	45.5%	5.669	0.129	NS

NS = Not Significant ($p > 0.05$).

Qualification: A statistically significant association was found between professional qualification and belief that proper IPC practices improve patient outcomes ($\chi^2 = 10.040$, $\mathbf{p = 0.007}$), with all BSN graduates (100%) agreeing compared to 80.0% of Post RN BSCN holders (Table 7). A near-significant trend was observed for the item "all HCWs/patients are potentially infectious" ($\chi^2 = 5.532$, $p = 0.063$).

Table 7: Association between professional qualification and key knowledge/practice responses using Pearson Chi-Square test. Statistically significant results ($p \leq 0.05$) are highlighted. Diploma=29, BSN=69, Post RN BSCN=5 (n=103)

Item	Diploma	BSN	Post RN	χ^2	p-value	Sig
All HCWs potentially infectious	82.8%	64.2%	100%	5.532	0.063	NS
Hand washing	82.8%	89.6%	100%	1.601	0.449	NS
Nosocomial definition	65.5%	73.1%	40.0%	2.676	0.262	NS
PPE available	89.7%	86.6%	60.0%	3.172	0.205	NS
IPC improves outcomes	96.6%	100%	80.0%	**10.040**	**0.007**	*

** Significant at $p < 0.05$. NS = Not Significant.*

5. DISCUSSION

The present study assessed knowledge, practices, and barriers related to infection prevention and control among 103 nurses at LUMHS tertiary care hospital. The findings reveal a pattern consistent with existing Pakistani literature: generally adequate knowledge but notable gaps in specific areas, combined with inconsistent practice implementation.

5.1 Knowledge-Practice Gap

The overall knowledge score (mean 3.61/5, with 60.2% demonstrating good knowledge) aligns with findings from Khowaja et al.[1] who reported satisfactory understanding among nurses at LIMHS Thatta, and Kousar et al.[3] who found high awareness at public sector hospitals in Gujrat. However, a critical knowledge gap was identified: 50.5% of nurses incorrectly believed that clean water alone is sufficient to destroy harmful microorganisms. This misconception is particularly concerning as it may directly impact hand hygiene quality and environmental disinfection practices. Similar knowledge gaps regarding specific disinfection mechanisms have been reported in studies from developing countries.[6,10]

The practice score (mean 7.31/9, 73.8% good practice) shows that while most nurses self-report adherence to infection control measures, over a quarter demonstrate suboptimal practice. This is consistent with Batool et al.[2] who found good overall practices but noted specific compliance gaps at tertiary care hospitals in Quetta. Similarly, Soomro et al.[4] reported that while nurses at Isra University Hospital demonstrated good basic knowledge, compliance with specific practices like contact and isolation precautions was only 56.7%.

The finding that 30.1% of nurses admitted to handling body fluids with bare hands when gloves are unavailable is particularly alarming. This echoes findings from Baqar et al.[5] who documented 33.3% non-compliance with glove use at Khairpur and noted that this places nurses at significant risk of occupational blood-borne infections including HBV, HCV, and HIV.

5.2 Barrier Analysis

The most significant contribution of this study is the systematic identification of institutional barriers. The three most prevalent barriers were: lack of annual IPC training (28.2%), inaccessibility of IC guidelines (25.2%), and self-reported inadequate knowledge (20.4%). These findings extend the work of Baqar et al.[5] who identified resource constraints and large patient numbers as barriers at Khairpur, and Khowaja et al.[1] who highlighted inadequate continuing education as a key obstacle.

The finding that 13.6% of nurses report PPE being unavailable creates an important contradiction with the high self-reported PPE usage rates. This suggests that when PPE is available, nurses use it, but supply inconsistency creates compliance gaps. This is a systemic institutional issue rather than an individual knowledge deficit, supporting the argument that interventions should target resource provision alongside education.[7,8]

5.3 Demographic Associations

Unlike Kousar et al.[3] and Baqar et al.[5] who found significant gender differences in infection control practices, the present study found no statistically significant gender associations (all $p > 0.05$). This may reflect that LUMHS, as a major teaching hospital, provides more standardized training across genders. Similarly, Batool et al.[2] found practices to be statistically non-significant according to age and work experience ($p > 0.05$) at Quetta tertiary hospitals.

The near-significant trend in self-assessed knowledge declining with age (95% for ≤25 vs. 62.5% for 31-40, $p = 0.088$) is noteworthy. This may reflect that younger, more recently trained nurses have greater confidence in their theoretical knowledge, or that curriculum improvements over time have enhanced training quality. However, this confidence may not translate to practice, as practice scores did not significantly differ by age.

The significant association between professional qualification and belief in IPC effectiveness ($p = 0.007$) is important. The finding that 20% of Post RN BSCN nurses did not agree that proper IPC improves outcomes is unexpected and warrants further investigation, though the small sample size (n=5) in this subgroup limits generalizability.

5.4 Comparison with International Literature

The knowledge-practice gap documented in this study is not unique to Pakistan. Studies from Nigeria,[11] Yemen,[12] Ethiopia,[6] and Bangladesh[13] have reported similar patterns. However, the magnitude of specific gaps-particularly the water disinfection misconception and bare-hand body fluid handling-appears more pronounced in resource-constrained settings, suggesting that resource availability moderates the relationship between knowledge and practice.

5.5 Limitations

This study has several limitations. First, the cross-sectional design prevents establishing causal relationships between barriers and practice. Second, self-reported data may be subject to social desirability bias, potentially inflating compliance estimates. Baqar et al.[5] demonstrated that observed compliance with hand washing (0.9%) was dramatically lower than self-reported rates-a limitation shared with the present study. Third, convenience sampling limits generalizability. Fourth, the single institutional setting means findings may not represent all tertiary hospitals in Pakistan. Future research should employ mixed-methods approaches combining surveys with observational data.

6. CONCLUSION

This study demonstrates that nurses at LUMHS tertiary care hospital possess generally adequate knowledge of infection prevention and control (60.2% good knowledge) and report reasonable adherence to standard practices (73.8% good practice). However, critical gaps persist: half of nurses hold a dangerous misconception about water disinfection, nearly a third handle body fluids without gloves when unavailable, and over a quarter lack access to annual IPC training.

The primary barriers identified are institutional in nature-lack of training frequency, guideline inaccessibility, and intermittent PPE supply-rather than individual knowledge deficits. These findings suggest that interventions targeting systemic resource provision and structured training programs would be more effective than additional knowledge-based education alone.

6.1 Recommendations

1. Mandate quarterly IPC training sessions rather than relying on annual workshops to address the 28.2% training gap
2. Ensure IC guidelines are printed and displayed at every nursing station to improve the 25.2% guideline accessibility deficit
3. Develop targeted educational modules specifically addressing the water disinfection misconception affecting 50.5% of nurses
4. Establish PPE supply monitoring and stock management systems to ensure consistent availability across all departments
5. Implement peer observation programs to bridge the gap between self-reported and actual compliance
6. Future research should include direct observational methods and use larger multi-center samples to improve generalizability

Conflict of Interest

The authors declare no conflicts of interest related to this study.

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Author Contributions

Tasleem: Conceptualization, methodology, data collection, data analysis, and manuscript writing.
Gobind Rai: Supervision, critical review, and final approval of the manuscript. All authors reviewed and approved the final version.

Data Availability

The datasets generated and analyzed during this study are available from the corresponding author upon reasonable request.

REFERENCES

1. Khowaja Z, Channar HB, Khowaja S, Channa S, Junejo Z. Knowledge and Practice of Infection Control Among Nurses at Liaquat Institute of Medical and Health Sciences Hospital, Thatta. *Journal of Health Rehabilitation Research.* 2024;4(3). doi:10.61919/jhrr.v4i3.1425
2. Batool S, Kousar S, Chandio K, Memon KN, Wilson NR. Infection Control Practices Among Nurses Working at Tertiary Care Hospitals of Quetta, Pakistan. *RADS Journal of Pharmacy and Allied Health Sciences.* 2023;1(1):1-8.
3. Kousar R, Afzal M, Parveen K, Hussain M, Gilani SA. Awareness and Practices of Nurses Related to Infection Control and Prevention at Public Sector, Pakistan. *Journal of Global Biosciences.* 2020;9(10):8025–8034.
4. Soomro A, Junejo Z, Soomro P, Soomro P, Bashir M. Knowledge and Practice of Standard Infection Control Precautions among Staff Nurses at Isra University Hospital, Hyderabad. *Pakistan Journal of Multidisciplinary Innovation.* 2024;3(2):1-8.
5. Baqar M, Kumar R, Khushk IA, Noonari AA, Ahmed F. Hospital Infection Control Practices among Nursing Staff Working at Tertiary Care Hospital of Khairpur, Sindh. *Journal of Liaquat University of Medical and Health Sciences.* 2018;17(02):117-122. doi:10.22442/jlumhs.181720563
6. Geberemariyam BS, Donka GM, Wordofa B. Assessment of knowledge and practices of healthcare workers towards infection prevention and associated factors in healthcare facilities of West Arsi District, Southeast Ethiopia. *Archives of Public Health.* 2018;76(1):69. doi:10.1186/s13690-018-0314-0
7. Haque M, Sartelli M, McKimm J, Bakar MA. Health care-associated infections–an overview. *Infection and Drug Resistance.* 2018;11:2321-2333.
8. Stewart S, Robertson C, Pan J, et al. Impact of healthcare-associated infection on length of stay. *Journal of Hospital Infection.* 2021;114:23-31.
9. World Health Organization. WHO Guidelines on Hand Hygiene in Health Care. Geneva: WHO; 2009.
10. O Nour M, Mohamed AM, Al Bishi HA, Amir AA. Knowledge, attitude and practices of healthcare providers towards MERS-CoV infection at Makkah hospitals, KSA. *International Research Journal of Medicine and Medical Sciences.* 2017;5(4):76-86.
11. Iliyasu G, Dayyab FM, Habib ZG, et al. Knowledge and practices of infection control among healthcare workers in a tertiary referral center in North-Western Nigeria. *Annals of African Medicine.* 2016;15(1):34-40.
12. Alrubaiee G, Baharom A, Shahar HK, Daud SM, Basaleem HO. Knowledge and practices of nurses regarding nosocomial infection control measures in private hospitals in Sana'a City, Yemen. *Safety in Health.* 2017;3(1):1-6.
13. Shahida SM, Islam A, Dey BR, Islam F, Venkatesh K, Goodman A. Hospital Acquired Infections in Low and Middle Income Countries: Root Cause Analysis and the Development of Infection Control Practices in Bangladesh. *Open Journal of Obstetrics and Gynecology.* 2016;6(1):28-39.
14. Gomarverdi S, Khatiban M, Bikmoradi A, Soltanian AR. Effects of a multi-component educational intervention on nurses' knowledge and adherence to standard precautions in intensive care units. *Journal of Infection Prevention.* 2019;20(2):83-90.
15. Alnajrani AH. Infection Control Across Healthcare Disciplines: Radiology, Nursing, Internal Medicine, Laboratory, and Optic Technicians. *Analysis and Metaphysics.* 2024;23:52-68.